

SPECIAL REPORT — APRIL 2026

The *Carbon* Question

Inside the **\$420 billion** industry that could determine whether the world meets its climate targets — and the engineers quietly rewriting its rules.

BY ELEANOR MARSH & DAVID KOWALSKI
GRID INTELLIGENCE REVIEW

ENERGY

DATA CENTERS

CLIMATE

847 TWh

+18.4%

\$420B

1.5°C



THE GRID PARADOX

How data centers became the world's hidden power grid

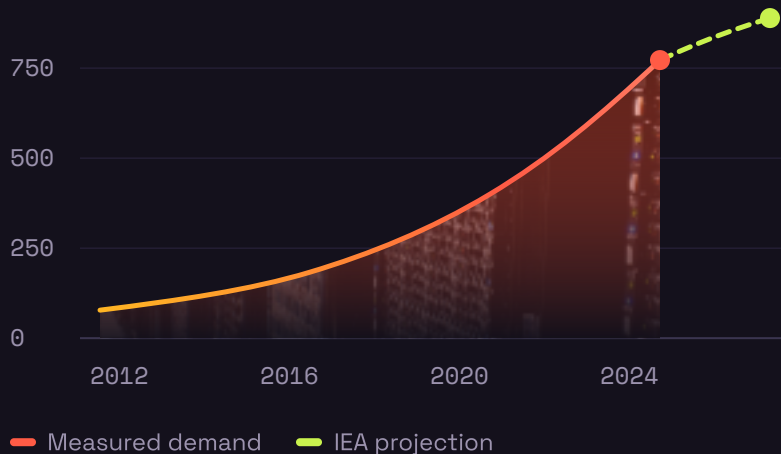
Twelve years ago, a [landmark paper in Nature Climate Change](#) estimated that the internet accounted for roughly 2% of global carbon emissions — comparable to aviation. Today that figure is contested by researchers at [Lawrence Berkeley National Laboratory](#), the [IEA](#) and the [Rocky Mountain Institute](#), who argue the real number is climbing fast and that conventional methodologies are **systematically underestimating** scope 3 emissions from hyperscale facilities.

HYPERSCALE

SCOPE 3

PUE

GLOBAL DATA-CENTER POWER DEMAND · TWH



Every efficiency gain the industry booked since 2015 was swallowed by the next generation of models inside eighteen months.

Marcus Feld · Uptime Institute



INSIDE THE MACHINE

POWER DRAW

12 MW

COOLANT / EXTERNAL

18.4°C / 26°C

"The servers never sleep. Neither do the engineers watching over them — in a facility that consumes as much electricity as a mid-size city."

COMPOUND DS-9 · NEVADA

Photo: BalticServers / Wikimedia Commons · CC BY-SA 3.0

Three figures that define the decade ahead

SHARE OF WORLD POWER

847_{TWh}

Estimated global data-center electricity consumption in 2025 — 3.2% of the world total.

DECISION HORIZON

1.5^{°C}

The Paris Agreement threshold — the ceiling the industry claims it can operate under. Contested by independent researchers.

VS. PRIOR 5-YR CYCLE

\$420_{BN}

Projected global investment in data-center infrastructure through 2030, driven by AI workloads.

OPERATIONAL METRICS

Real-time facility performance

POWER USAGE EFFECTIVENESS

1.18

RENEWABLE COVERAGE

74%

CARBON INTENSITY

128 g/kWh

WATER USAGE EFFECTIVENESS

0.82

UPTIME SLA

99.999%

COMMITMENTS VS. REALITY

Industry pledges audit

- ✓ 100% renewable energy by 2030 — pledged by 6 of the top 10 operators
- ✓ Net-zero scope 1 & 2 emissions across owned facilities
- ✗ Scope 3 supply-chain emissions reporting — fewer than 12% disclose
- ✗ Verifiable third-party audits of water consumption claims
- ✓ Cooling efficiency targets — PUE < 1.2 for all new builds
- ✗ Binding commitments on embodied carbon in hardware procurement

REGIONAL BREAKDOWN

Data-center electricity consumption by region, 2025

REGION	TWH	YOY	RENEWABLE	RELATIVE DEMAND
North America	312	+18.4%	41%	
Asia Pacific	276	+24.1%	28%	
Europe	148	+9.7%	68%	
Middle East & Africa	62	+31.8%	12%	
Latin America	49	+14.2%	54%	

SOURCES: IEA DATA CENTRE ENERGY CONSUMPTION MONITOR 2025 · ROCKY MOUNTAIN INSTITUTE · SYNERGY RESEARCH GROUP. RENEWABLE SHARE INCLUDES PPAS AND RECS.

PHOTOGRAPHY: WIKIMEDIA COMMONS — BALTICSERVERS (CC BY-SA 3.0) · SCOTT BEALE / LAUGHING SQUID (CC BY 2.0) · HANS HILLEWAERT (CC BY-SA 4.0).

We keep calling this an energy crisis. It isn't.
It's a *priority crisis* — and right now the
market has decided that milliseconds matter
more than megatons.



— Jane Thornton • Senior Energy Analyst, Grid Intelligence Review



FROM LAND TO LIVE

How a hyperscale facility reaches production

01

Site Selection

Grid capacity, water access, seismic risk, fiber routes and tax incentives evaluated across 40+ candidate zones.

6–18 MONTHS

02

Power Negotiations

PPAs with utilities, grid interconnect studies, generator permitting and renewable certificate procurement.

12–24 MONTHS

03

Construction

Civil works, structural steel, raised floors, chilled-water or direct-air cooling, redundant fiber pathways.

18–30 MONTHS

04

Commissioning

Load-bank testing, DCIM integration, failover drills, thermal validation, security audits, compliance sign-off.

3–6 MONTHS

05

Operations

24/7 NOC monitoring, predictive maintenance, capacity planning and continuous PUE optimisation toward <1.15.

ONGOING

METHODOLOGY & NOTES

Sources, caveats & conventions

Consumption figures draw on data published by the [International Energy Agency](#), cross-referenced with the [Uptime Institute Global Data Center Survey 2025](#) and corroborated by metered readings from [Open Infrastructure Map](#). Where sources diverge we apply the conservative estimate and flag the discrepancy.

Carbon intensity figures are **location-marginal**, not **market-average** — a distinction accounting for roughly 40% of reported variance between operators claiming identical renewable targets.

All PUE calculations follow
$$\text{pue} = \frac{\text{total_facility_power}}{\text{it_equipment_power}}$$
 as standardised by the Green Grid consortium. To reproduce the dataset locally press  then .

Scope 3 emission boundaries follow the GHG Protocol Corporate Standard.¹ Water usage effectiveness is measured at the facility boundary, not the source-water withdrawal point.²

¹ GHG PROTOCOL CORPORATE STANDARD (2015).

² GREEN GRID WHITE PAPER #49: WUE (2016, UPD. 2024).

TERMS & TRADEMARKS

What the labels do and don't mean

"The absence of a universal standard is not a gap in knowledge — it is a gap in political will."

Dr. Amara Osei-Bonsu · Director of Sustainable Infrastructure, MIT Energy Initiative

GHG PROTOCOL

ISO 50001

EN 50600

ASHRAE A2

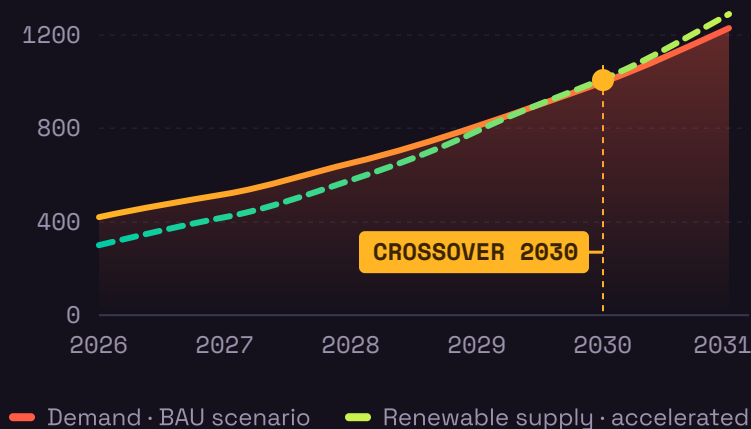
TIER IV

Power Usage Effectiveness™ and PUE™ are trademarks of The Green Grid®. LEED® is a registered trademark of the U.S. Green Building Council.

Prepared in accordance with GRI® Energy Standard 302. All figures are pre-publication estimates; audited data releases Q3 2026.

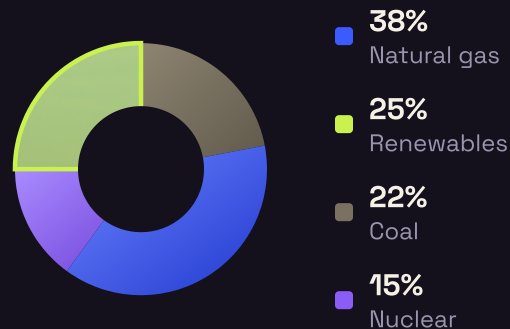
THE OUTLOOK

Projected demand vs. renewable supply



ENERGY MIX, 2025

Where the power comes from



The numbers are unambiguous. The question that remains is not scientific — it is political.

SVG rendering test — vector marks at scale



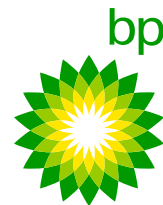
shell.svg

150 × 139

ExxonMobil

exxon-mobil.svg

250 × 47



bp-helios.svg

283 × 377

أرامكو السعودية
saudi aramco

aramco.svg

596 × 168



sinopec.svg

82 × 85 mm



PetroChina

petrochina.svg 307 × 325 · REBUILT



ExxonMobil

أرامكو السعودية
saudi aramco

<blockquote> variants — dispatch test

01 • BARE — <blockquote>text</blockquote>

Milliseconds are cheap. Megatons are not.

03 • BLOCK CHILD — <blockquote><p>...</p></blockquote>

Every efficiency gain was spent within eighteen months.

05 • FOOTER + CITE — <blockquote><p>...<footer><cite>

We priced latency and forgot to price the grid.

— *Uptime Institute, 2025*

02 • CITE ATTR — <blockquote cite="...">

Boundaries decide the answer before measurement begins.

04 • NESTED — <blockquote><blockquote>...

The gap is not in knowledge.

06 • INLINE STYLE — <blockquote style="...">

| *Attribution is a layout problem, not a data problem.*